Natural Build-Up

3. Originality and natural build-up of features on top of the base mechanics (simplicity):

WARNING! Some fine reads regarding slot ideas and creativity follow below.

> Complexity has nothing to do with intelligence, simplicity does.

Actually you can google these words and read more about them. There are many interpretations out there, and many people discuss what exactly they were meant to say, and many of them disagree with them having their own arguments. Personally, I think they are as accurately said as possible, but not everyone has the experience to understand their meaning. Let me try to present my point of view and I assume the point of view of their author as well. But first... think about them a little and try to interpret them on your own. How did you manage to interpret them? Feel free to express your opinion in the comments of this topic.

What I think is that their author is trying to say that our "complexities" are affected by the search for simplicity. To get from A to B, there is movement. First state there is tension. Second state, movement. Final stage is *simply* stationary. Most of us can make it complex, but can you make it simple and solve the same problem? Thats where you have to involve intelligence. **Simplicity is the key. Complexity alone can betray a lack of idea.** Do you agree?

Any topic that challenges creativity, art, and originality of ideas is difficult to describe, especially if that creativity hides a lot of mathematics and combinatorics behind it, but for this topic we have the exact examples that we will use. The examples include 3 slot games:

- One that is the bearer of a BRILLIANT, yet simple idea.
- Another that tries to replace that same brilliant idea and thus solve a problem, but fails and derails, yet results in another into something beautiful in itself, but not well configured.
- And a third that manages to replicate the idea of the first game in a different way and does it much better.

So lets begin.

First we will need to get to know some payout mechanics in depth:

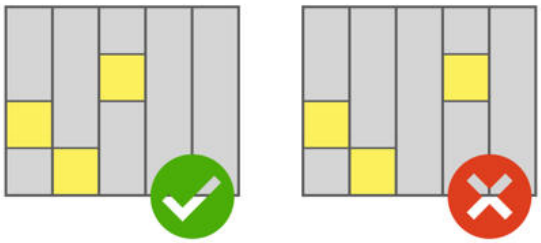
WAYS (or also known as ALL WAYS)

With WAYS payout strategy, **Wins are paid when matching symbols appear anywhere on adjacent reels from left to right starting from the leftmost reel.** This practically means that we include all possible simple lines and for a screen with size N reels and M rows we will have exactly M^N different payout matches. For example, for a screen of size Reels x Rows = 5 x 4, we will have total 1024 simple paylines.

MEGA WAYS

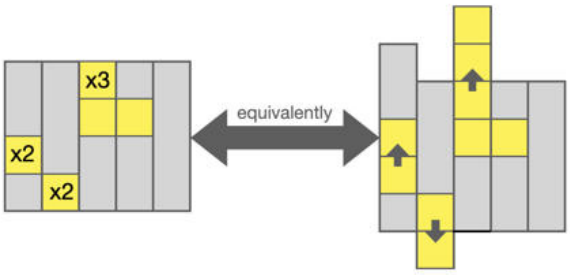
The **Megaways™** slot payout mechanic is the most played mechanic ever. It was developed by Big Time Gaming (BTG). They **patented** this innovative system, which allows for a dynamic number of paylines that can vary with each spin, providing a unique gaming experience. The mechanism has been **licensed** to various other game developers, allowing them to create their own Megaways slots using this popular format.

Key Features of Megaways™ Payout Strategy:



WAYS

Special multipliers called **WAYS MULTIPLIERS** are defined particularly for this payout strategy. Those multipliers are decorators above the original symbols on which they are positioned. A way multiplier increases the **number of symbol occurrences** on the reel of the screen and in that way makes the symbol with the multiplier stronger in the ways calculation. Here is a visual example of a WAY MULTIPLIER and its interpretation:



WAY MULTIPLIERS

While quite interesting and innovative, these multipliers create events that build on the mechanics synthetically and are not part of it. For this reason, you create events that are difficult to track and for this reason such games can hardly be certified in a region like Pennsylvania, as the max odds regulations (or at least those before 2024-11-12) restrict them.

1. Dynamic Paylines:

- Instead of fixed paylines, Megaways™ games use a "ways to win" system.
- Each reel can display a different number of symbols (usually between 2 and 7 symbols per reel).
- The total number of ways to win is calculated by multiplying the number of symbols on each reel. For example, if the reels display 3, 4, 5, 6, 5, and 4 symbols respectively, the total ways to win would be 3 × 4 × 5 × 6 × 5 × 4 = 7,200 ways.

More

2. High Volatility:

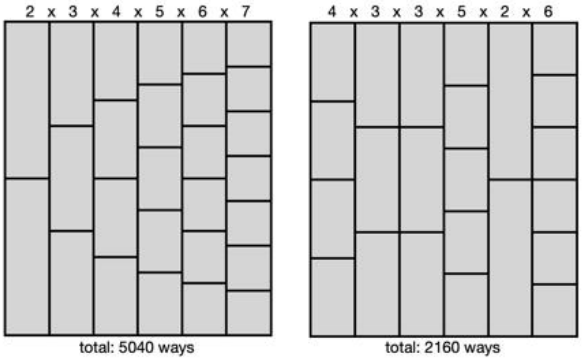
- Megaways™ games are known for their high volatility, meaning wins may be less frequent but potentially larger when they occur.
- The dynamic nature of the paylines and the cascading reels contribute to this volatility.

3. Cascading Reels:

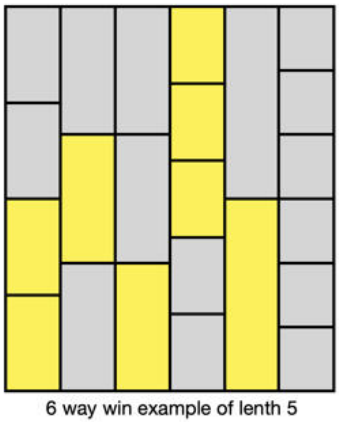
- Many Megaways™ games feature cascading or tumbling reels.
- When a winning combination occurs, the winning symbols disappear, and new symbols fall into place, potentially creating additional wins in a single spin.

Advantages of Megaways™ Payout Strategy:

- **High Engagement:** The constantly changing number of ways to win keeps players engaged.
- **Massive Win Potential:** The high volatility and multipliers can lead to significant payouts.
- **Innovative Gameplay:** The dynamic nature of the mechanic sets it apart from traditional slot games.



MEGA WAYS CONFIGURATIONS



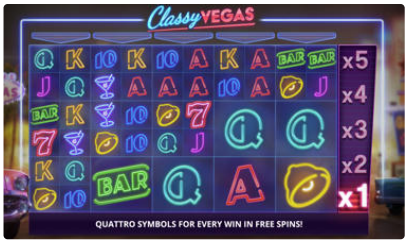
EXAMPLE OF 6 WAY WIN

Notice how the WAY MULTIPLIERS that build on the natural WAY mechanics from the previous observed pay strategy now are already built into the MEGAWAYS mechanics and are part of it, rather than being synthetic events that build on it. This way, the megaways mechanics are not subject to any regulations. It's a brilliant idea that simultaneously injects a lot of variation, solves problems, and is beautiful. The idea behind this basic payout mechanic is to simply expand it with a feature functionality in it, which is a simple but not trivial idea.

The first game that we will observe is the Big Time Gaming's game **Bonanza Megaways** released in 2016. Bonanza was the **first-ever MegaWays slot**, introducing the groundbreaking mechanic to the world. It features a gold mining theme with 6 reels and up to 117,649 ways to win. The game includes free spins with unlimited win multipliers, making it highly volatile but potentially very rewarding. Its simplicity, combined with the innovative mechanic, made it an instant classic. The Legacy of Bonanza is that it set the standard for all future MegaWays games and remains a fan favorite.

Let's imagine that we want to create a mechanic close to Megaways and thus extract its beneficial properties, but without having to buy the license for this mechanic. In fact, Cayetano's design team have tried to do something similar before with the game Classy Vegas (without pots) released in 2018-05-15 (2 years after the first Megaways game). Here is what they did:

Classy Vegas 01



Classy Vegas 02

Classy Vegas 03

Did they manage to extract the Megaways beneficial properties? How do you think?

One of the most important features of the Megaways basic payout mechanic is related to volatility. It can explode with a big win stake multiplier at any time even if you are not in Free Spins bonus and this is due to the variability that many ways can inject. In how many ways can a player be paid in Classy Vegas? Well, lets calculate them. The calculation of all ways to win can be seen in the table on the right. We can see that **1,098 << 117,649** and we can conclude that the design team wasn't even close to replicating the mechanics of Megaways and certainly didn't manage to extract its beneficial properties. Still, they managed to do something different and innovative, but too complicated and failed to solve the task they set themselves. Admirations for the attempt, though. Once again simplicity proves to be more difficult than complexity and certainly much more profitable than it.

So the first slot game that we explore was the Bonanza Megaways, which carried within itself the brilliant idea of its mechanics combining WAYS with WAYS MULTIPLUERS and calling it MEGAWAYS.

The second slot game that we explore was. Classy Vegas, which we proved that was not very successful in terms of replication and leading to a completely different result, which is unfortunately too sophisticated and almost impossible to configure well, due to its synthetic mechanics.

Now lets explore the third slot game which does more than well in its attempt to inherit the winning characteristics of the Megaways mechanics in a natural way! This game is presented below and is a typical example of a good and simple idea.

Classy Vegas: f(x, y) = f(x-1, y-1) + f(x-1, y+1)

0	0	0	0	0	0	0	0	0	0
1	1	2	3	6	10	19	33	61	108
1	2	3	6	10	19	33	61	108	197
1	2	4	7	13	23	42	75	136	244
1	2	4	7	13	23	42	75	136	244
1	2	3	6	10	19	33	61	108	197
1	1	2	3	6	10	19	33	61	108
0	0	0	0	0	0	0	0	0	0
Total ways to win:					1098				

CV total ways to win

So here it is, **Napoleon 2** Fat Stacks slot developed by Blueprint Gaming and released in 2023-11-23. So the particular Napoleonic War theme game has a basic WAYS payout strategy with a screen 6 x 4 and respectively 4,096 ways to win. What's interesting here is what feature they managed to add to the regular Ways mechanics so that they get the beneficial properties of Megaways games without having to pay for a license and at the same time build on the basic mechanics in a natural and easily understandable way.

The idea is simple and it hides in the so called Fat Stacks:

Fat Stacks

If you land two or three identical single symbols on the reels, they can consolidate into double or triple symbols (Fat Stacks), creating even more ways to win at these slots. **Any symbol can be subject to the Fat Stacks feature, with the exception of wilds and scatters.**

FAT STACKS

WAYS TO WIN FAT STACKS

You can check the game feeling here: [Napoleon 2 Fat Stacks DEMO LINK](#).

Without the other features, this game would have the potential to pass even Pennsylvania regulations, as only multiples of 2 and 3 and their combinations should be tracked. But anyway, the combination requirement was dropped with the regulation update from [PGCB 2024-11-12](#). What we want to emphasize here is the original and simple idea that achieves the goals described above.